

Functional Annotation Report: *paaK* (PP_3279, UniProt Q88HS4)

Phenylacetate–Coenzyme A Ligase of *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

paaK (ordered locus PP_3279; UniProt Q88HS4) of *Pseudomonas putida* KT2440 encodes **phenylacetate–coenzyme A ligase (phenylacetyl-CoA ligase, EC 6.2.1.30)**, an ATP-dependent (AMP-forming) acyl-CoA synthetase of the **ANL/adenylate-forming (AMP-dependent synthetase/ligase) superfamily**. It catalyzes the **first, committed step of the aerobic phenylacetate (PA) catabolic pathway**, activating the chemically inert aromatic carboxylic acid phenylacetate to its coenzyme A thioester:



The enzyme is a **cytoplasmic**, soluble ~48-kDa monomer that acts on intracellular phenylacetate. Its product, phenylacetyl-CoA, is the **central convergent intermediate** (the "phenylacetyl-CoA catabolon") into which many peripheral aromatic pathways (phenylalanine, 2-phenylethylamine, styrene, *n*-phenylalkanoic acids, environmental aromatics) funnel; PA-CoA is then processed by a distinct ring-epoxidizing/CoA-dependent route to acetyl-CoA and succinyl-CoA for entry into central metabolism. PA-CoA is also the physiological inducer of the pathway.

The gene identity is **unambiguous and fully consistent with the UniProt annotation**: symbol *paaK*, EC 6.2.1.30, phenylacetyl-CoA ligase family, AMP-dependent synthetase/ligase domains (IPR000873, IPR028154, IPR045851, IPR042099), in *P. putida* KT2440.

2. Gene/Protein Identity Verification

Attribute	UniProt Q88HS4	Literature consistency
Symbol	<i>paaK</i> (also <i>pha</i> nomenclature in KT2440; <i>pcl</i> in <i>P. putida</i> U)	✓ consistent; <i>paaK</i> is the standard PA-CoA ligase symbol (E. coli, Thermus, Burkholderia)
EC number	6.2.1.30	✓ matches purified enzyme reaction
Family	Phenylacetyl-CoA ligase family / ANL superfamily	✓ AMP-forming acyl-CoA synthetase fold
Organism	<i>P. putida</i> KT2440	✓ genome study P 12534466 maps the <i>pha</i> cluster
Reaction	PA + ATP + CoA → PA-CoA + AMP + PPi	✓ P 2324116 P 18414813

Conclusion: This is the correct, well-characterized gene. No ambiguity; the symbol, EC number, family, domains, and organism are mutually consistent with an extensive primary literature. (Note: the KT2440 phenylacetate cluster was originally annotated with *pha* gene names by Jiménez et al. 2002 [PMID 12534466]; the community-standard *paa/paaK* nomenclature, used by UniProt, is equivalent.)

3. Primary Molecular Function — The Catalyzed Reaction

3.1 Reaction and mechanism

PaaK is an **AMP-forming ligase** of the ANL (Acyl-CoA synthetase / NRPS adenylation / Luciferase) superfamily. Like all members, it works by a **two-step, adenylate-forming mechanism**: 1. **Adenylation**: phenylacetate + ATP → phenylacetyl-adenylate (PA-AMP) + PPi. 2. **Thioesterification**: PA-AMP + CoA → phenylacetyl-CoA + AMP.

The domain architecture reported for Q88HS4 — a large N-terminal adenylation domain (ANL_N; IPR042099) plus a small mobile C-terminal domain (AMP-binding C; IPR045851/ IPR028154) — is the structural hallmark of this superfamily, in which the C-terminal domain rotates (~140°) between an "adenylation" and a "thioester-forming" conformation ("domain alternation") to catalyze the two half-reactions.

3.2 Direct biochemical evidence (purified enzyme)

The enzyme was first purified to homogeneity (513-fold) from *Pseudomonas putida* and biochemically characterized (Martínez-Blanco et al. 1990,):

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- **Reaction requirements:** ATP, CoA, phenylacetate, and Mg^{2+} (as $MgCl_2$). - **Size:** single $\sim 48 \pm 1$ kDa polypeptide (monomer) by SDS-PAGE. - **Optimum:** pH 8.2, 30 °C. - **Apparent K_m :** ATP 9.7 mM, CoA 1.0 mM, phenylacetate 16.5 mM. - **Inhibition:** by Cu^{2+} , Zn^{2+} , Hg^{2+} and by thiol-reactive reagents (N-ethylmaleimide, DTNB, p-chloromercuribenzoate) — indicating a functionally important cysteine/thiol.

A sensitive coupled fluorometric assay (adenylate kinase / pyruvate kinase / lactate dehydrogenase, monitoring NADH) confirmed the AMP-forming stoichiometry (Rodríguez-Aparicio et al. 1991,).

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The homologous PaaK from *Thermus thermophilus* provided a clean in-vitro demonstration of the exact reaction and higher substrate affinities: - **PA + CoA + MgATP $\rightarrow$ PA-CoA + AMP + MgPPi**, V_{max} 24 $\mu\text{mol}/\text{min}/\text{mg}$, K_m ATP 6 μM , CoA 30 μM , PA 50 μM (Erb et al. 2008,).

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3.3 Gene cloning confirms function

The gene encoding this enzyme (*pcl* in *P. putida* U, ortholog of KT2440 *paaK*) was cloned by Tn5 mutagenesis and shown to encode the 48-kDa PA-CoA ligase; heterologous expression in *E. coli* conferred PA-CoA ligase activity and the ability to grow on phenylacetate as sole carbon source (Miñambres et al. 1996,).

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In *E. coli* W, *paaK* was identified within the *paaABCDEFGHIJK* catabolic operon as the gene

"coding for a phenylacetyl-CoA ligase that catalyzes the activation of PA to phenylacetyl-CoA" (Ferrández et al. 1998,).

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4. Substrate Specificity

PaaK is **highly specific for unsubstituted phenylacetate**: - The purified *P. putida* enzyme **does not use** 3-hydroxy-, 4-hydroxy-, or 3,4-dihydroxyphenylacetate as substrates, and shows a specificity **distinct from other acyl-CoA-activating enzymes**; hydroxy-/methyl-phenylacetyl derivatives and short/long-chain fatty acids (>C5), acetate, butyrate and succinate neither serve as substrates nor induce the enzyme (Martínez-Blanco et al. 1990,).

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- *T. thermophilus* PaaK is **highly specific toward phenylacetate**, with only low activity on 4-hydroxyphenylacetate (Erb et al. 2008,).

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Comparative structural/kinetic work in *Burkholderia cenocepacia* (which carries two ligases, PaaK1 and PaaK2) rationalizes this specificity: the specific enzyme has a compact binding pocket, whereas the more promiscuous PaaK1 has a **larger binding pocket** that can accommodate hydroxylated PA derivatives (3-OHPA, 4-OHPA) (Imolorhe & Cardona 2011,).

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P. putida KT2440's PaaK corresponds to the strict phenylacetate-specific type.

Interpretation: Narrow specificity ensures PaaK channels only genuine phenylacetate — the hub metabolite of the catabolon — into the CoA-dependent pathway, avoiding wasteful activation of related aromatics that belong to other routes.

5. Subcellular Localization / Where It Acts

- PaaK is a **soluble, cytoplasmic** enzyme. It is a monomeric globular protein with no signal peptide, membrane-spanning region, or secretion signal; the purified protein behaves as a

soluble 48-kDa species

(
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).

- It acts on the **intracellular pool of phenylacetate**. Phenylacetate first enters the cell via a dedicated, energy-dependent **phenylacetic acid transport system (PATs)** (active transport, $K_m \approx 13 \mu\text{M}$, driven by the membrane energization/electron-transport state), after which cytoplasmic PaaK activates it (Schleissner et al. 1994,

P 8002592).

Thus the enzyme performs its catalytic function **in the cytoplasm**, at the interface between substrate uptake and downstream CoA-dependent ring processing.

6. Pathway Context and Biological Process

6.1 The phenylacetyl-CoA "catabolon"

PaaK is the **committed entry enzyme** of the aerobic phenylacetate degradation pathway — a **CoA-dependent "hybrid" pathway** that is fundamentally different from classical aerobic aromatic catabolism (which uses ring-hydroxylating dioxygenases on free aromatic acids). This pathway operates in ~16% of sequenced bacteria, including *E. coli* and *P. putida* (Teufel et al. 2010,

P 20660314).

In KT2440, genomic analysis identified the **phenylacetate (*pha/paa*) pathway** as one of four central aromatic catabolic routes (alongside the protocatechuate and catechol β -ketoadipate branches and the homogentisate pathway) (Jiménez et al. 2002,

P 12534466).

Multiple **peripheral (upper) pathways converge on phenylacetyl-CoA**, making PaaK the funnel point for: - **L-phenylalanine** (via phenylpyruvate/phenylacetaldehyde → phenylacetate), - **2-phenylethylamine** (via the *mao*/amine oxidase route), - **styrene** (via the *sty* upper pathway,

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- **n-phenylalkanoic acids** and other environmental aromatics/pollutants.

6.2 Steps downstream of PaaK (fate of the product PA-CoA)

Once PaaK forms phenylacetyl-CoA, the ring is attacked while activated as a thioester (Teufel et al. 2010,

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2012,

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1. **PaaABC(E)** — a multicomponent, **di-iron monooxygenase/epoxidase** — converts PA-CoA to **ring-1,2-epoxyphenylacetyl-CoA** (and can deoxygenate it back, an intrinsic detox mechanism). 2. The epoxide isomerizes to a seven-membered O-heterocyclic enol ether, **oxepin-CoA**. 3. **PaaZ** (bifunctional (R)-enoyl-CoA hydratase + NADP⁺ aldehyde dehydrogenase) hydrolytically cleaves the ring to **3-oxo-5,6-dehydrosuberyl-CoA**. 4. **β -oxidation-like steps** (PaaFGHIJ homologs) degrade this to **acetyl-CoA and succinyl-CoA**, which enter the TCA cycle.

6.3 Regulation

- The product **PA-CoA is the true physiological inducer** of the pathway: it antagonizes the **PaaX** transcriptional repressor, de-repressing the catabolic operon (Ferrández et al. 1998,

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This places PaaK's product at the center of pathway induction.

- In KT2440, phenylacetate metabolism is under **carbon catabolite repression (CCR)** by glucose/gluconate, acting at the transcriptional level. The intracellular signal is **2-keto-3-deoxy-6-phosphogluconate (KDPG)**: an *edd* mutant (no KDPG) shows high phenylacetyl-CoA ligase activity even in glucose+PAA (CCR abolished), whereas wild type is repressed (Kim et al. 2009,

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PA-CoA ligase activity and *paaK* transcription track pathway induction across conditions (O'Leary et al. 2002,

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Kim et al. 2007,

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6.4 Biotechnological/physiological relevance

Because PaaK generates phenylacetyl-CoA, expressing the *P. putida* ligase gene in *Penicillium chrysogenum* increased the pool of this direct precursor and **boosted benzylpenicillin (penicillin G) biosynthesis ~1.8–2.2-fold** (Miñambres et al. 1996,

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— direct evidence that the enzyme's product is phenylacetyl-CoA and that PaaK is the rate-relevant activating step.

7. Supported and Refuted Hypotheses

Supported: - H1: PP_3279/paaK encodes phenylacetate-CoA ligase (EC 6.2.1.30). ✓ (multiple biochemical + genetic lines). - H2: The enzyme catalyzes $\text{PA} + \text{ATP} + \text{CoA} \rightarrow \text{PA-CoA} + \text{AMP} + \text{PPi}$, Mg^{2+} -dependent, AMP-forming. ✓ - H3: It is strictly specific for unsubstituted phenylacetate. ✓ - H4: It is cytoplasmic and acts on imported phenylacetate. ✓ - H5: It is the committed first step of the CoA-dependent catabolon converging on $\text{PA-CoA} \rightarrow \text{acetyl-CoA} + \text{succinyl-CoA}$. ✓ - H6: It is induced by phenylacetate and subject to KDPG-mediated CCR. ✓

Refuted / Not supported: - That PaaK acts on hydroxylated phenylacetates (3-/4-OH-PA) in *P. putida* — **refuted** for the specific enzyme (no activity;

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- That the aromatic ring is opened on the free acid by a dioxygenase (classical model) — **refuted**; ring processing occurs on the CoA thioester via epoxidation (

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8. Limitations and Future Directions

1. **Direct structural data** for the *P. putida* KT2440 PaaK (Q88HS4) itself have not been reported; mechanistic/structural inferences draw on homologs (*Thermus*, *Burkholderia*, *E. coli*) and the ANL superfamily. An experimental crystal/cryo-EM structure of the KT2440 enzyme with PA-AMP/CoA would confirm active-site residues and the domain-alternation mechanism.
 2. Most **quantitative kinetics** derive from *P. putida* U and thermophilic/other homologs; strain-KT2440-specific kinetic constants and any second paralog should be verified from the KT2440 genome.
 3. In-vivo flux control by PaaK versus the transporter (PATs) and the downstream oxygenase under different carbon regimes merits systems-level study.
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9. Key References

- Martínez-Blanco et al. 1990, *J. Biol. Chem.* — purification/characterization of PA-CoA ligase (PMID **2324116**).
- Rodríguez-Aparicio et al. 1991 — sensitive assay confirming AMP-forming reaction (PMID **2009287**).
- Schleissner et al. 1994 — phenylacetate active transport system (PMID **8002592**).
- Miñambres et al. 1996 — cloning of *pcl/paaK*; penicillin precursor boost (PMID **8969218**).
- Ferrández et al. 1998 — *paa* cluster, *paaK* function, PA-CoA as inducer (PMID **9748275**).
- Jiménez et al. 2002 — KT2440 aromatic catabolism genomics; *pha* pathway (PMID **12534466**).
- O'Leary et al. 2002 — *paaK* transcription linked to styrene/PA growth (PMID **11959447**).
- Erb et al. 2008 — recombinant PaaK reaction and kinetics (PMID **18414813**).
- Kim et al. 2009 — KDPG-mediated CCR of PA metabolism/ligase (PMID **19406896**).
- Teufel et al. 2010 — elucidation of the CoA-dependent phenylalanine/phenylacetate pathway (PMID **20660314**); 2011 (PMID **21296885**); 2012 (PMID **22398448**).
- Imolorhe & Cardona 2011 — two PaaK ligases, binding-pocket/specificity (PMID **22919580**).

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